THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 10: **Light: Shadows and Reflections** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Materials through which light passes partially are: 1

(a) translucent **(b)** transparent **(c)** luminous **(d)** opaque

2. The image in a pinhole camera is formed on a: 1

(a) the front **(b)** the hole

(c) the lens **(d)** screen at the back

3. The angle between the incident ray and the normal is the angle of: 1

(a) reflection **(b)** incidence **(c)** deviation **(d)** refraction

4. A periscope uses: 1

(a) no mirrors **(b)** one prism only

(c) two plane mirrors **(d)** two lenses only

5. Materials through which light does not pass are: 1

(a) luminous **(b)** translucent **(c)** transparent **(d)** opaque

6. In a pinhole camera, the image becomes sharper when the hole is: 1

(a) large **(b)** absent **(c)** covered **(d)** small

7. The angle between the reflected ray and the normal is the angle of: 1

(a) reflection **(b)** deviation **(c)** incidence **(d)** refraction

8. A kaleidoscope forms patterns by: 1

(a) refraction **(b)** multiple reflections

(c) absorption **(d)** dispersion

9. A shadow is formed when: 1

(a) light passes through glass **(b)** light is bent

(c) an opaque object blocks light **(d)** light is reflected

10. The image in a pinhole camera is: 1

(a) virtual **(b)** erect

(c) of same size always **(d)** real

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. What is the difference between a real and a virtual image? 2

12. State the laws of reflection. 2

13. Why is AMBULANCE written in reverse? 2

14. Name two uses of plane mirrors. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Draw (describe) the reflection of a ray at a plane mirror with the incident angle 50°. 3

16. Explain why we cannot see our image in a wall but we can in a mirror. 3

17. What are the uses of a periscope? 3

OR

List three properties of the image in a plane mirror. 3

18. Describe how a kaleidoscope works. 3

OR

If the angle between the incident ray and the mirror is 30°, find the angles of incidence and reflection. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the construction and working of a periscope. 5

OR

Explain with an activity how to find the position of the image in a plane mirror. 5

20. Explain regular and diffused reflection with examples and show how the quality of the surface matters. 5

OR

Write short notes on transparent, translucent and opaque materials with uses. 5

END OF QUESTION PAPER — Total: 40 Marks

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